EV Charger Gaps in New Jersey

Hadia Khan

Abstract

New Jersey's transportation sector accounts for 39% of the State's net greenhouse gas emissions, making it the largest emissions sector in the State and a critical climate policy issue. Transportation electrification has been a focus of recent New Jersey legislation which includes EV charging infrastructure development to facilitate EV adoption. This study identifies DCFC charging gaps in rural areas of New Jersey and L2 and DCFC charging gaps in urban areas of New Jersey to help policymakers, utilities, and the private sector to design and effectuate EV charging incentives.

Introduction and Background

As of 2022, New Jersey's transportation sector accounts for 39% of the State's net greenhouse gas ("GHG") emissions, making it the largest emissions sector in the State and a critical climate policy issue (NJDEP, 2025). Electric vehicles ("EVs") emit less GHG emissions than a conventional internal combustion engine vehicle, making them a great alternative for mitigating GHG emissions (Alternative Fuels Data Center, n.d.) Transportation electrification has been a focus of recent New Jersey legislation such as The Electric Vehicle Act of 2020 (2019) which directed the Board of Public Utilities to advance the adoption of EVs and development of EV charging infrastructure. EVs and EV chargers can also serve as grid assets: by employing managed charging and integrating them as an energy source/storage, EVs can abate utility costs and improve grid resilience by reducing load demand and stress (evenergy, 2025).

To facilitate EV adoption and meet State goals of 2 million EVs on the road by 2030, the State requires a robust charging network. Having a scaled public charging network in particular, fills gaps of other charging options (i.e. residents without access to residential chargers) and reduced range anxiety, further building consumer confidence in EVs (National Renewable Energy Laboratory, 2023). This public network is estimated to need 76,479 Level 2 ("L2") charger ports and 3,918 Direct Current Fast Charger ("DCFC") ports by 2030 (National Laboratory of the Rockies, n.d.) As of November 5, 2026, New Jersey has 4,451 L2 ports and 2,074 DCFC ports, meaning the private sector has been able to meet about 52.9% of the

DCFC charger demand but only 5.8% of the L2 charger demand (EValuate, 2026). This leaves a major gap to be filled.

Identifying where EV charger gaps exist will inform EV charging policy and boost EV adoption. This study identifies DCFC charging gaps in rural areas of New Jersey and L2 and DCFC charging gaps in urban areas of New Jersey. Current charger geographic distribution is exhibited in *Figures 1-3*. Rural areas are anticipated to mainly utilize residential chargers and DCFCs rather than L2s, whereas urban areas will require a robust network of L2s along with DCFC. This study will help identify what locations could benefit the most from EV charger incentives.

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Dimension:     XY
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Geodetic CRS:  NAD83
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Projected CRS: WGS 84 / Pseudo-Mercator
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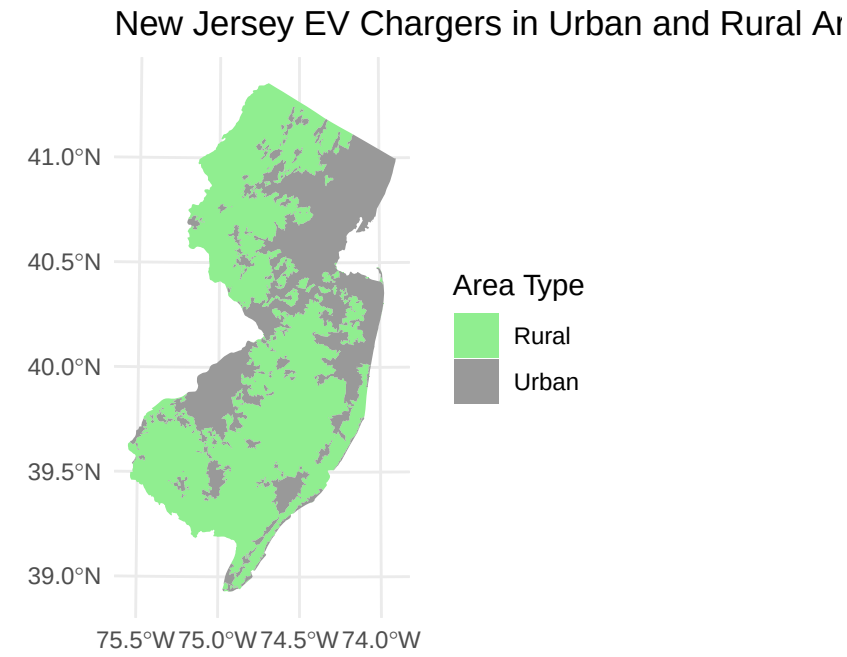
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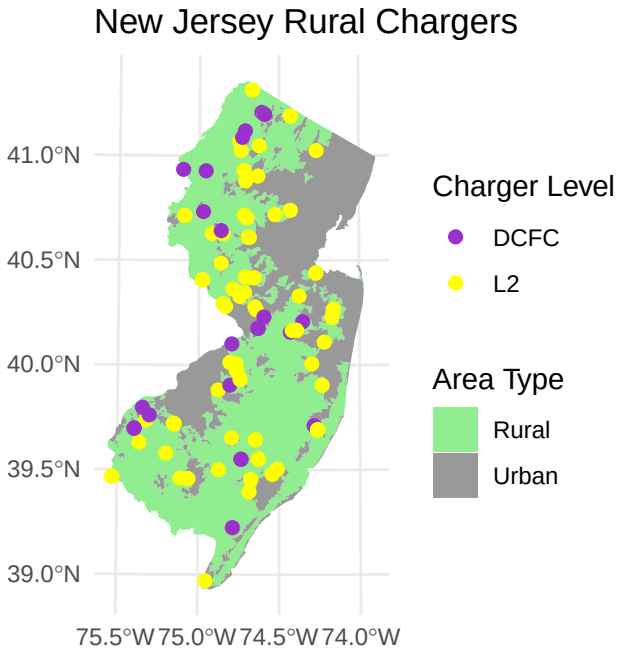
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Charger Level
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Area Type
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Figure 2: New Jersey Urban L2 Chargers

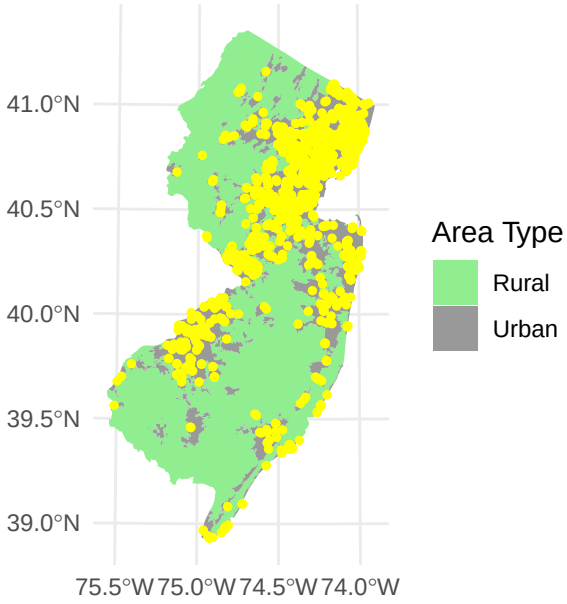
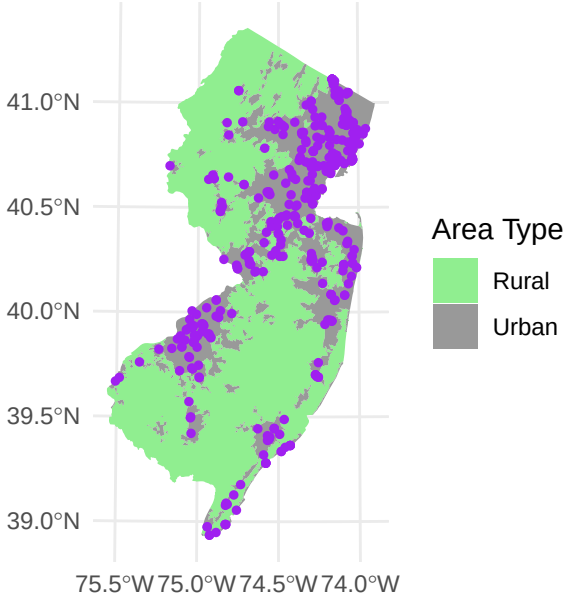


Figure 3: New Jersey Urban DCFC Chargers



Methodology

Analysis

This study contains two types of analyses: 1) geospatial parametric gap identification, and 2) Local Moran's I hotspot analyses.

For the purposes of this study, urban areas are identified according to the USA Census Urban Areas and rural areas are those that do not meet those criteria. To qualify as a USA Census Urban Areas, census blocks must have at least 2,000 housing units or have a population of at least 5,000.

For rural DCFC gap identification, there are two parameters:

1. Corridor-based gaps: Rural municipalities within 3 miles of a major corridor where there are no DCFC chargers within 15 miles.
2. Population-based gaps: Rural municipalities with a population of at least 5,000 where there are no DCFC chargers within 15 miles.

For urban DCFC gap identification, a Local Moran's I hotspot analysis was conducted on a municipaltiy level. For urban L2 gap identification, a Local Moran's I hotspot analysis was conducted on a census block level. Local Moran's I hotspot analysis identifies hotspots or coldspots which are regions with significantly more or less EV chargers than its neighbors–with a 90% confidence level in this study. High-low outliers are census blocks/municipalities with a high amount of EV chargers but surrounded by census blocks/municipalities with a low amount of EV chargers. Low-high outliers are census blocks/municipalities with a low amount of EV chargers but surrounded by census blocks/municipalities with a high amount of EV chargers.

Data

EV charging stations were used from the Alternative Fuels Data Center Electric Vehicle Charging Station Locations tool and accessed August 26, 2026. USA Census Urban Areas (2020), major corridors (Primary and Secondary Roads), and New Jersey census blocks were accessed from the United States Census Bureau's 2026 TIGER/Line Shapefiles. New Jersey municipalities were accessed from the New Jersey Office of GIS's Municipal Boundaries of NJ, Hosted, 3857 (2025).

Results

The rural gap identification analysis finds that there are only two rural municipalities that meet the criteria of being within a corridor-based DCFC gap: Shiloh and Stow Creek (*Figure 4*). This finding suggests that rural areas in New Jersey do not suffer from a large DCFC gap which would inhibit EV adoption and EV chargers in rural areas need not be overincentivised.

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Projected CRS: NAD83 / New Jersey (ftUS)

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Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: 59037.62 ymin: 10682.7 xmax: 194613.4 ymax: 280250.2
Projected CRS: NAD83 / New Jersey

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Geometry type: POINT
Dimension:     XY
Bounding box:  xmin: 61244.57 ymin: 10930.67 xmax: 197814 ymax: 275428.3
Projected CRS: NAD83 / New Jersey


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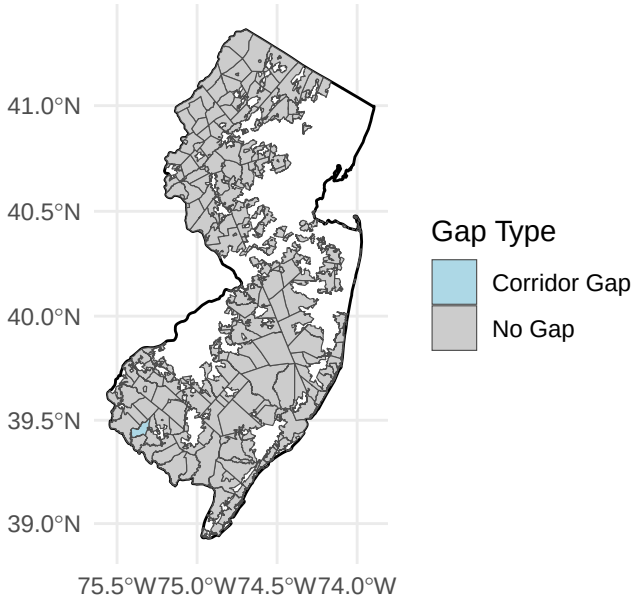
Figure 4: Rural Municipality DCFC Gaps



A preliminary chloropleth map of urban DCFCs (*Figure 5*) shows that high DCFC concentration is aligning with New Jersey population hubs which serves the goal of providing charger access to where it is needed. The preliminary chloropleth map of urban L2 chargers (*Figure 7*) supports

evidence that there is a major gap in L2 chargers across the State's urban areas with several census block outliers.

The urban DCFC hotspot analysis finds that there are fourteen high-low outlier census blocks (Howell, Ewing, Plainsboro, Elizabeth, Kearny, Rutherford, Waldwick, Allendale, Hillside, Newark, Jersey City, Absecon, Harrison, and Secaucus), as well as one low-high outlier: Northvale (*Figure 6*). The latter implies a potential gap to be filled but further research is necessary to verify if there is a need in that location.

```
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Dimension:     XY
Bounding box:  xmin: 60116.6 ymin: 10737.02 xmax: 200270.4 ymax: 280253.2
Projected CRS: NAD83 / New Jersey
```
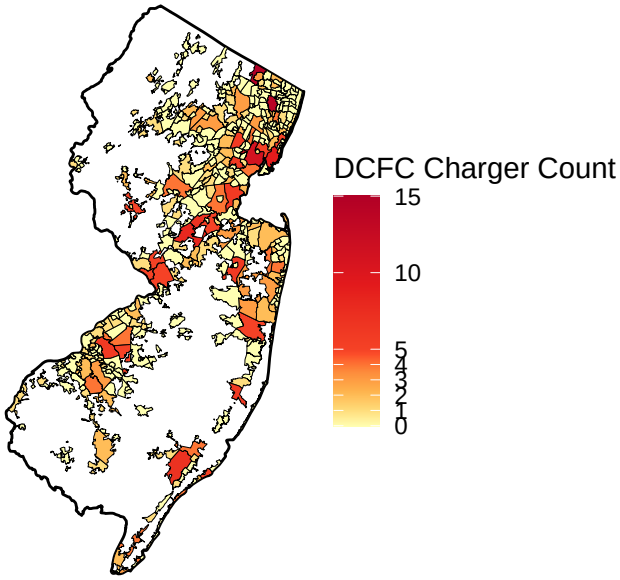

Figure 5: Urban DCFC Chargers by Municipality

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Projected CRS: WGS 84 / Pseudo-Mercator
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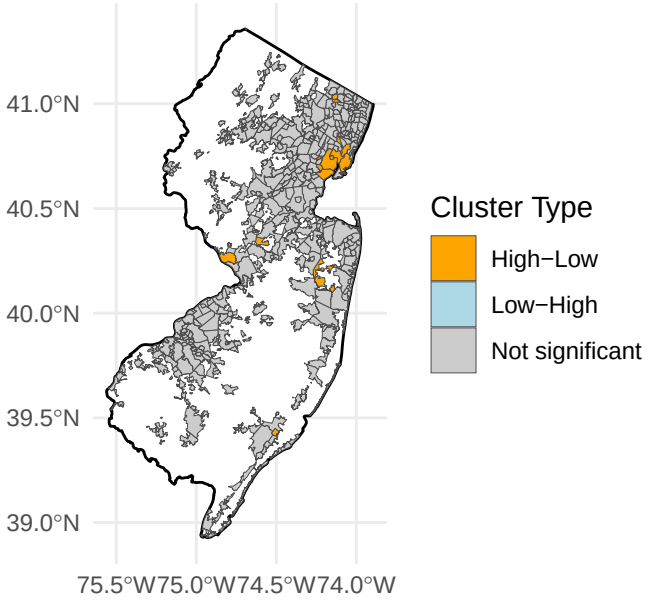
Figure 6: Urban DCFC Hot Spot Analysis by Munic



The urban L2 hotspot analysis finds that there are [] high-low outlier census blocks (*Figure 8*). This confirms there is a great need for L2 charger development across the State to meet EV adoption needs.

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Geodetic CRS:  NAD83
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Simple feature collection with 2645 features and 6 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -170.7893 ymin: -14.36542 xmax: 145.7916 ymax: 64.92889
Geodetic CRS:  NAD83
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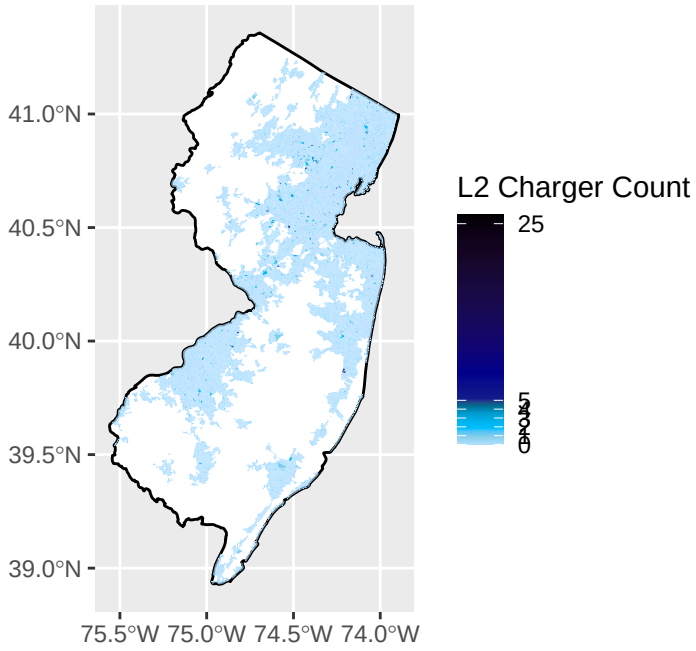


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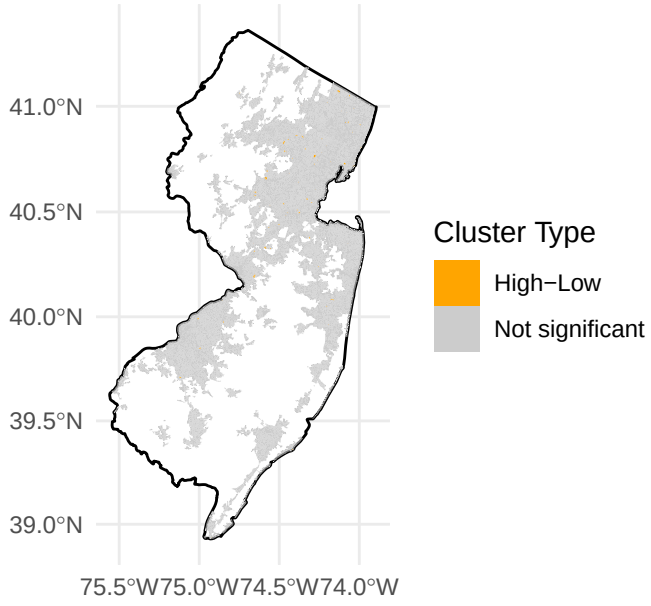
Figure 8: Urban L2 Hot Spot Analysis by Census E



Recommendations

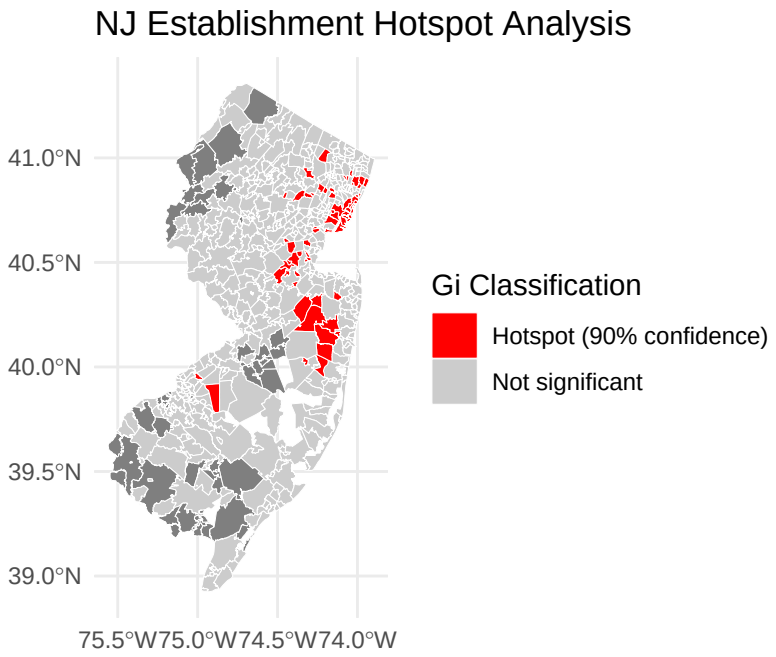
The main EV charging gaps in New Jersey are rural DCFC and urban L2 charger build-out. Soltanpour et al. (2025) found that doubling EV charging investment in L2 chargers and DCFCs reduces unserved EVs by 61%. This study can help policymakers narrow down where DCFC incentives should be prioritized in rural locations. In addition, this study confirms the need for L2 charger build-out across the state.

The New York State Energy Research and Development Authority and Transportation and Climate Initiative propose that siting accessible EV chargers includes factors such as visibility, length of stay, proximity to building entrances, pedestrian and vehicle traffic, parking and weather conditions, and more (WXY Architecture + Urban Design, 2012). State incentives for urban L2 chargers should consider establishment concentrations (*Appendix A*) and walkability (e.g. USEPA's National Walkability Index, 2026) to determine optimal locations for L2 utilization on a local level. In places with an abundance of establishments, EV owners may be able to park and charge their car within walking distance while they work, run errands, see sites, etc. Siting of EV chargers to optimize utilizaation is critical in the initial adoption phase of EVs as slow uptake can incur significant demand charges which impact both the consumer and the utility (Muratori et al., 2019).

Overall, access to a robust DCFC and L2 charger network quells concerns about EV range and convenience for New Jersey residents driving long distances and those lacking residential charging, building confidence in EV adoption.

Appendix A

```
Simple feature collection with 1 feature and 9 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -75.55954 ymin: 38.92852 xmax: -73.89363 ymax: 41.35742
Geodetic CRS:  NAD83
  STATEFP  STATENS     GEOIDFQ GEOID STUSPS       NAME LSAD       ALAND
1      34 01779795 0400000US34    34     NJ New Jersey   00 19049229866
      AWATER                       geometry
1 3533374440 MULTIPOLYGON (((-74.0422 40...
```



References

Alternative Fuels Data Center. (n.d.) *Station Locator.* Retrieved August 26, 2026, from https://afdc.energy.gov/stations#/find/nearest?fuel=ELEC

Alternative Fuels Data Center. (n.d.) *Emissions from Electric Vehicles.* Retrieved November 5, 2026, from https://afdc.energy.gov/vehicles/electric-emissions

Atlas Public Policy. (n.d.) *EValuateNJ.* Retrieved November 5, 2026, from https://atlaspolicy.com/evaluatenj/

Electric Vehicle Act of 2020 ("EV Act"), N.J.S.A. § 48:25-1 et seq. (2020).

evenergy. (2025). *The Utility Playbook: Turning EV Grid Risk into a $30 Billion Opportunity.* https://platform.ev.energy/resources/value-of-managed-charging

Muratori, M., Kontou, E., & Eichman, J. (2019). Electricity Rates for Electric Vehicle Direct Current Fast Charging in the United States, *Renewable and Sustainable Energy Reviews, 113.* https://www.osti.gov/pages/servlets/purl/1543251

National Laboratory of the Rockies. (n.d.) EVI-X: Electric Vehicle Infrastructure Toolbox. Retrieved November 5, 2026, from https://afdc.energy.gov/evi-x-toolbox#/evi-pro-ports

National Renewable Energy Laboratory. (2023). *The 2030 National Charging Network: Estimating U.S. Light-Duty Demand for Electric Vehicle Charging Infrastructure.* https://afdc.energy.gov/files/u/publication/2030-charging-network.pdf

New Jersey Department of Environmental Protection. (April 2025). *New Jersey Greenhouse Gas: 2025 Mid-Cycle Update Report.* https://dep.nj.gov/wp-content/uploads/ghg/ghg-inventory-mcu-2025-1.pdf

New Jersey Office of GIS. (10 April 2025). *Municipal Boundaries of NJ, Hosted, 3857.* Retrieved October 14, 2026, from https://njogis-newjersey.opendata.arcgis.com/datasets/newjersey::municipal-boundaries-of-nj-hosted-3857

Soltanpour, A., Rostami, A., Ghafarnezhad, B., Ghamami, M., & Zockaie, A. (2025). Strategic deployment of Level-2 charging infrastructure at long-distance trip endpoints integrated with fast charging network to enhance EV tourism, *Sustainable Cities and Society, 130.* 10.1016/j.scs.2025.106516

United Census Bureau. (2023). *CB2300CBPAll Sectors: County Business Patterns, including ZIP Code Business Patterns, by Legal Form of Organization and Employment Size Class for the U.S., States, and Selected Geographies: 2023.* https://data.census.gov/table/CBP2023.CB2300CBP

United States Census Bureau. (n.d.) TIGER/Line® Shapefiles. Retrieved October 14, 2026, from https://www.census.gov/cgi-bin/geo/shapefiles/index.php

United States Environmental Protection Agency. (8 January 2026). *National Walkability Index User Guide and Methodology.* https://www.epa.gov/smartgrowth/national-walkability-index-user-guide-and-methodology

WXY Architecture + Urban Design. (November 2012). *Siting and Design Guidelines for Electric Vehicle Supply Equipment.* New York State Energy Research and Development Authority and Transportation and Climate Initiative. https://www.transportationandclimate.org/sites/default/files/EV_Siting_and_Design_Guidelines.pdf